Assignment

Objective 1:

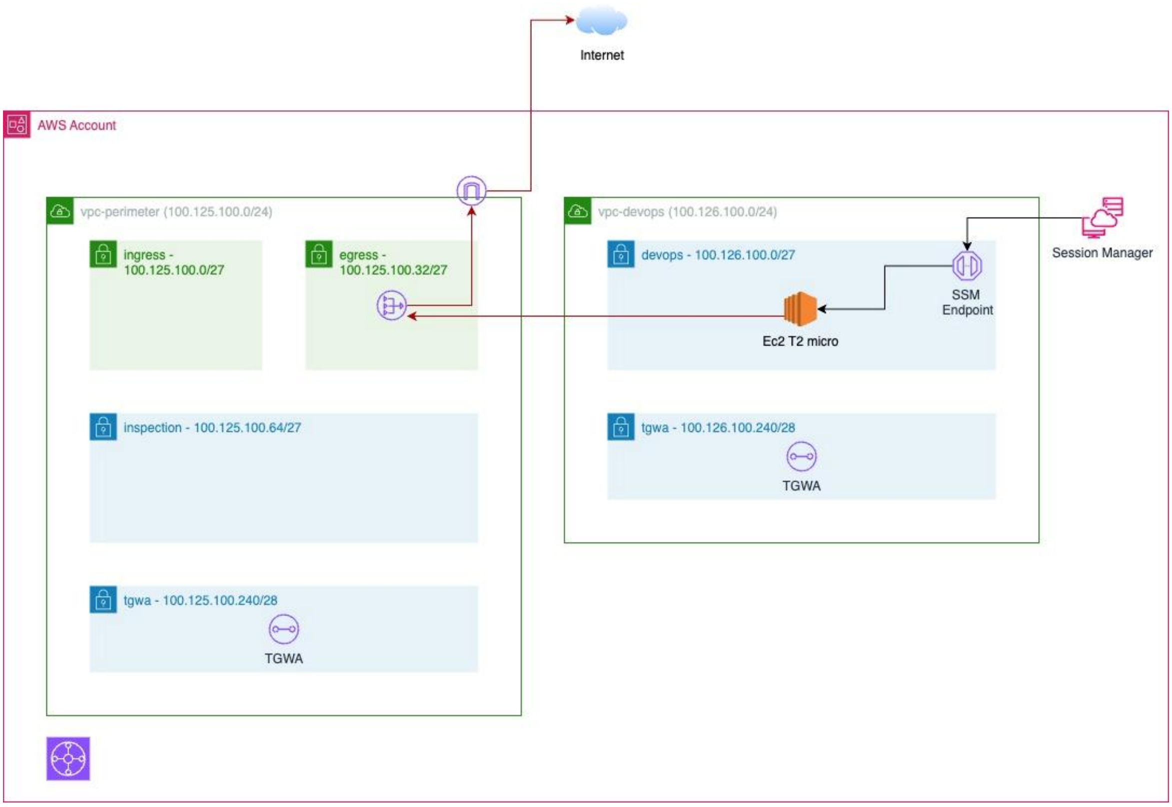
Your task is to provision a Private DevOps Virtual Machine (VM) within a private DevOps Virtual Private Cloud (VPC).

This setup should allow the DevOps VM to:

1. **Outgoing HTTPS Traffic:** Access the internet to download necessary libraries via central egress inside perimeter VPC.

Additionally, DevOps engineers should be able to SSH into the DevOps VM using the AWS Systems Manager (SSM) Session Manager with SSM private endpoint (not via public AWS egress)

Requirements:
Use Infrastructure as Code (IaC) with Terraform to provision and configure all necessary cloud resources as per below architecture by referencing page below terraform folder structure only please write your own terraform module.



Notes:

- VPCs, S3 for tfstate and IAM user for tf-user can create manually (not require iac)
- Use Terraform module instead of Resource Block directly. (Reference below terraform folder structure)
- Require individual route table and individual nacl for each subnet. (write as nested module inside subnet module.

Objective 2:

Your task is to automate above IAC deployment using gitlab pipeline. In Objective 1, all cloud resources are provision directly from your local machine (aws cli and terraform) now via gitlab shared runner (need aws cli and terraform inside)

Requirements:

- Use Gitlab pipeline to deploy above IAC deployment via shared runner.
- Require two pipeline for each vpc (perimeter-network-layer-pipeline, perimeter-service-layer-pipeline, devops-network-layer-pipeline, devops-service-layer-pipeline)

Notes:

You may not require to do terraform import as its use same tfstate file inside S3. Objective of this task is to automate via gitlab pipeline instead from local machine.

Expected folder structure in gitlab

```
DevOps assignment (main repository)
|- Training1 (folder)
  |-Perimeter (folder)
  |-DevOps (folder)
  |-Modules (folder)
  |-.gitlab-ci.yml (yaml file)
```

Outcome:

DevOps engineer push IAC code into Gitlab and Gitlab pipeline trigger to provision all cloud resources via gitlab shared runner.